American Computer Science League

2021-2022 ● Contest 1: Short Problems ● Junior Division

1. Computer Number Systems

Convert 2122_{10} to octal.

A. 4211
B. 4121
C. 1224
D. 4112
E. 2114

2. Computer Number Systems

The Fibonacci sequence {0, 1, 1, 2, 3, 5, 8, 13, …} is formed by adding the two previous terms: 5 + 8 = 13. Which number in the sequence when converted to binary would include the 4th occurrence of the substring "10" when considering each number separately?

A. 8
B. 13
C. 21
D. 34
E. 55

3. Recursive Functions

Find $f(10)$, given

$$f(x) = \begin{cases} f(x-2) + 3 & \text{if } x \geq 7 \\ f(x-1) - 1 & \text{if } 3 < x < 7 \\ x^2 & \text{if } x \leq 3 \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 12

4. Recursive Functions

Find $f(20, 12)$, given

$$f(x,y) = \begin{cases} f(x-3,\ y+1) + 3 & \text{if } x > y \\ 2 * f(x+2,\ y-3) - 1 & \text{if } x = y \\ x + y & \text{if } x < y \end{cases}$$

A. 29
B. 34
C. 63
D. 23
E. 60

5. What Does This Program Do? (Branching)

What is the output when this program is executed?

```
a = 0 : b = 2 : c = 4 : d = 6
if c > d / b then
   c = b * d
end if
if b ^ 2 < c / d then
  c = b ^ 2
else
   b = b ^ 2
end if
if c - d < a * b then
   a = b + c
end if
if a ^ 2 > 1 then
   b = b + a
else
   b = b + d
end if
output c / d - b + a ^ 3 * d
```

A. -8
B. 0
C. -2
D. -4
E. 2

美国计算机科学联赛

2021-2022 ● 第一轮: 短问题 ● 初级组

1. 计算机数字系统

请将 2122_{10} 转化为八进制。

A. 4211
B. 4121
C. 1224
D. 4112
E. 2114

2. 计算机数字系统

斐波那契数列 {0, 1, 1, 2, 3, 5, 8, 13, …} 由前两项相加而成，比如：5 + 8 = 13。将数列中的每个数字转换为二进制后，第四次出现子串 “10” 的是哪个数？

A. 8
B. 13
C. 21
D. 34
E. 55

3. 递归函数

根据以下条件，求 $f(10)$。

$$f(x) = \begin{cases} f(x-2) + 3 & x \geq 7 \\ f(x-1) - 1 & 3 < x < 7 \\ x^2 & x \leq 3 \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 12

4. 递归函数

根据以下条件，求 $f(20, 12)$。

$$f(x,y) = \begin{cases} f(x-3, y+1) + 3 & x > y \\ 2 * f(x+2, y-3) - 1 & x = y \\ x + y & x < y \end{cases}$$

A. 29
B. 34
C. 63
D. 23
E. 60

5. 此程序是怎么运行的？（分支）

执行这个程序会打印输出什么？

```
a = 0 : b = 2 : c = 4 : d = 6
if c > d / b then
   c = b * d
end if
if b ^ 2 < c / d then
  c = b ^ 2
else
   b = b ^ 2
end if
if c - d < a * b then
   a = b + c
end if
if a ^ 2 > 1 then
   b = b + a
else
   b = b + d
end if
output c / d - b + a ^ 3 * d
```

A. -8
B. 0
C. -2
D. -4
E. 2